

Omkar Patil

📍 Tempe, AZ ✉ opatil3@asu.edu 🔗 omkarpatil18.github.io 🔗 LinkedIn 📄 Google Scholar

Education

Arizona State University

August 2023 — August 2027

Doctor of Philosophy in Computer Science (3.9/4.0)

Indian Institute of Technology, Madras

August 2015 — May 2020

Bachelor's in Mechanical Engineering and Master's in Robotics and AI (8.22/10)

Publications

Self-Supervised Keyframe Discovery for Horizon-Invariant Behavior Cloning [🔗](#) December 2026

NeurIPS 2026 / Prabin Kumar Rath, **Omkar Patil**, Nakul Gopalan

Self-supervised method that discovers information-critical observations (mnemonics) as keyframes, conditioning behavior cloning policies on them to retain context over long horizons, without annotations.

You've Got a Golden Ticket: Improving Generative Robot Policies With A Single Noise Vector [🔗](#) March 2026

arXiv / **Omkar Patil**, Ondrej Biza, Thomas Weng, Karl Schmeckpeper, Wil Thomason, Xiaohan Zhang, Kausik Sivakumar, Robin Walters, Nakul Gopalan, Sebastian Castro, Stephen Hart, Eric Rosen

Search method to find a single constant noise vector (golden ticket) that improves pretrained diffusion/flow matching robot policies including VLAs (GR00T N1.5, $\pi 0$, SmolVLA) without any retraining.

StageCraft: Execution Aware Mitigation of Distractor and Obstruction Failures in VLA Models [🔗](#) October 2026

IROS 2026 / Kartikay Pangaonkar*, Prabin Rath*, **Omkar Patil***, Nakul Gopalan

Training-free VLM-based scene staging for VLAs ($\pi 0.5$, SmolVLA); +40% absolute success on real robots.

Factorizing Diffusion Policies for Observation Modality Prioritization [🔗](#) May 2026

ICRA 2026 / **Omkar Patil**, Prabin Rath, Kartikay Pangaonkar, Eric Rosen, Nakul Gopalan

Method to prioritize observational modalities, such as vision over tactile, for learning diffusion policies.

Experience

Graduate Research Assistant, Logos Robotics Lab, Arizona State University August 2023 — Present

PhD research on robot learning for real-world manipulation with Dr. Nakul Gopalan.

- Build ROS 2 teleoperation pipelines (leader arm, VR headset) for a Franka Research 3 arm and a ROBOTIS AI Worker semi-humanoid, used to collect 500+ demonstrations across 10 tasks.
- Build data-collection and real-time inference pipelines to deploy trained policies, including Diffusion Policy, ACT, and VLAs ($\pi 0.5$, SmolVLA), on real robots.
- Fine-tune policies with behavior cloning and on-robot policy improvement, and deploy simulation-trained policies on hardware (sim-to-real) for research experiments and demonstrations.

Research Intern, RAI Institute July 2025 — January 2026

Post-training of VLAs and generative robot policies on downstream reward functions.

- Developed a latent-steering-based approach that improved generative policies on 46 of 51 tasks without retraining, resulting in the filing of a patent and a publication: <https://lottery-tickets.rai-inst.com/>.
- Demonstrated that derivative-free episodic policy improvement can be competitive with policy-gradient and actor-critic methods, especially for VLAs, potentially with a 15 $\times$ reduction in wall-clock time.

Senior NLP Quantitative Researcher, Wells Fargo August 2020 — July 2023

- Developed a novel metric to evaluate the quality of LLM-generated paraphrases (arXiv [🔗](#), patent [🔗](#)).
- Surveyed and built LLM-enabled document generation, speeding up internal documentation (arXiv [🔗](#)).

Service

Reviewer: CoRL 2025, 2026; ICRA 2025, 2026; IROS 2025, 2026

Organizer: Compositional and Modular Learning in the Era of Scaling in Robotics (IROS 2026) [🔗](#)

Skills

Robot Learning: Imitation Learning, Behavior Cloning, VLA Post-training (Latent Steering, Policy Improvement), Generative Policies (Diffusion, Flow Matching, ACT), Reinforcement Learning (PPO, SAC, Offline RL, On-robot RL, Sim-to-Real)

Robotics: Motion Planning (IK, Trajectory Optimization), Perception (Vision, Point Clouds), Control, Hardware (Franka FR3, ROBOTIS AI Worker), Teleoperation (Leader Arm, VR), ROS 2, MuJoCo, Isaac Sim

Tooling: VLAs ($\pi 0.5$, SmolVLA, GR00T N1.5), LeRobot, PyTorch, JAX, Hugging Face, W&B, Docker, Git